

Rahul Tammalla

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Professional Summary

AI Software Engineer building production agentic large language model (LLM) systems and RAG pipelines, along with the APIs and cloud backends on AWS that put them in front of real users. Strong software-engineering fundamentals in Python, FastAPI, Docker, and CI/CD, with a focus on reliable, low-latency products, backed by M.S.-level research on LLM evaluation and human preference modeling.

Education

Clarkson University

M.S., Applied Data Science, Merit Scholar, GPA: 3.8/4.0 (Graduate TA, Probability & Statistics)

Amrita Vishwa Vidyapeetham

Bachelor's in Computer Science and Engineering

Potsdam, NY

Aug 2024 – May 2026

Kerala, IND

May 2017 – Jun 2021

Skills

LLM & Generative AI: RAG, Multi-Agent Systems, MCP, Prompt Engineering, Fine-Tuning, LangChain, LangGraph, Hugging Face Transformers, PyTorch, Mistral, Claude, Gemini, Groq, Structured Output, Agentic Reasoning, LLM Evaluation

Software & API Engineering: Python, FastAPI, Django, REST APIs, Async I/O, Microservices, System Design

Retrieval & Data: FAISS, Chroma, mem0, Semantic Search, Hybrid Retrieval, Embeddings (GTE, HuggingFace), SQL, Pandas, PySpark

Cloud & DevOps: Docker, Kubernetes, AWS (S3, Glue, Lambda, EMR), CI/CD, Airflow, Git

Machine Learning: scikit-learn, XGBoost, LightGBM, Model Evaluation

Data Engineering: Snowflake, Redshift

Work Experience

BalanX Bio

AI Developer Intern

East Lansing, MI

Jul 2026 – Present

- Built a long-term emotion-memory system in **Python** for a **multi-modal LLM** avatar, wiring **MCP** tool integrations and **agentic reasoning** over a self-hosted **mem0/Chroma** vector store to cut false-positive writes **40%**.
- Designed a **hybrid retrieval** scorer combining **semantic search** over **HuggingFace** embeddings with recency decay on **Groq**-hosted LLMs, running **LLM evaluation** across **5** approaches and **4** models to lift relevance **20%**.

Clarkson University

AI/ML Research Assistant

Potsdam, NY

Jan 2025 – Jun 2026

- Engineered a statistical and **ML** pipeline over **20M+** NEMSIS emergency records (hypothesis testing, **70+** engineered features), identifying 15 statistically significant predictors associated with stroke-related dispatch delays.
- Built scalable **Python/SQL/PySpark** pipelines that resolved **1M+** missing values to **95%** data quality via multivariate imputation, then trained **Random Forest**, **XGBoost**, **LightGBM**, **SVM** models via **scikit-learn** that improved prediction accuracy by **20%**.
- Built an **NLP** extraction pipeline with **spaCy**, BeautifulSoup, and regex that structured unstructured biological research text into **300+** normalized entities, boosting downstream data quality **45%**. Contributed feature development and QA testing for **ERDPlus**, a web-based database-modeling tool.

I4DI (Institute for Development Impact)

AI Engineer Intern

Washington, DC

May 2025 – Aug 2025

- Built a **multi-agent** survey-generation system with **Mistral 7B** and **Claude** on **LangChain**, using **Django** backend services and **JSON-schema** validation to take survey authoring from brief to delivery, cutting manual effort by **90%**.
- Owned **system design** for a **RAG** pipeline using **FAISS** with a real-time knowledge-base override, deployed in **Docker** containers returning answers under **3 seconds**; batched **PyTorch GTE** embeddings cut latency **42%** and lifted **Precision@K 30%**.

Cognizant Technology Solutions

Data Engineer

Remote

Oct 2021 – Aug 2024

- Built and maintained production **Airflow/SQL** ETL pipelines on **Snowflake/Redshift** and **AWS** (S3, Glue, EMR, Lambda) supporting **Gilead Life Sciences'** enterprise analytics platform, processing **5M+** daily records while reducing pipeline failures **70%** and improving query performance **30%**.
- Cut manual reporting effort **30%** across **3+** source systems by building **5+ Tableau/Power BI** dashboards on **CI/CD** pipelines (**AWS/Kubernetes**) that sustained **95%+** uptime for finance and operations reporting.

Projects

Agentic AI – DevOps Agent Swarm

GitHub

- Architected a 6-agent autonomous monitoring system on **LangGraph** with **Gemini**-powered root-cause reasoning, exposing **FastAPI** orchestration endpoints across **Docker**-containerized **microservices** with **async** event processing, configurable escalation policies, and durable audit logs.

NLP – Clinical Consent NER & Risk Classifier

GitHub

- Extracted 5 structured entity types from unstructured clinical consent documents using **spaCy** NER and **Transformers**-based fine-tuning, reaching **85% F1** across all categories and enabling automated consent-language risk classification.

Certifications & Achievements

Certificate of Completion: Claude 101, Anthropic | Jun 2026 • AI Fluency Framework & Foundations, Anthropic | Jun 2026